



UTM
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Task : Exploring MS Azure & Power BI

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Group : 1

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1.0 Project Overview

This project aims to address business problems where a business company does not have a clear picture of customer demographics, and how this would affect the sales of the product. Although a huge volume of customer and sales information is stored on an on-premises SQL Server database, they cannot be easily analyzed and used to make decisions.

The main objective of this project is to examine the demographics of the customers, especially the gender and the titles of the customers and learn about their influence on the product sales in various categories. Stakeholders also need a KPI dashboard that will give them critical information like total sales revenue, total products sold, customer distribution, by gender. Moreover, the dashboard should have interactive filtering so that the users can view data by product category and gender, with a user-friendly interface for data-based queries.

This project will involve the building of an entire data engineering pipeline on the Azure platform, including data extraction to visualization. The project can be applied to the real-world business context because it shows how organizations could convert raw data into valuable insights to aid data-driven decision-making and enhance business performance.

2.0 Description of Solution

The proposed solution is to have an automated data engineering pipeline, which would receive data that is on-premise and transform it into a cloud-based analytics system. The solution is based on a clear workflow of data ingestion, transformation, loading, reporting and monitoring.

2.1 Data Ingestion

Azure Data Factory (ADF) is used to extract data out of on-premises SQL Server databases. It allows the integration of data reliably and aids in scheduling. The result of the extraction is subsequently loaded to Azure Data Lake Storage (ADLS Gen2) which serves as the central location in the pipeline.

2.2 Data Transformation

Azure Databricks is then used to process the data. Layered style is employed with Bronze, Silver and Gold layers.

- **Bronze Layer** stores raw data
- **Silver Layer** cleans and transforms the data
- **Gold Layer** stores final, business-ready data

This approach improves data quality and makes the data easier to use for analysis.

2.3 Data Loading and Reporting

After transformation, the final data is loaded into Azure Synapse Analytics. Structured views are created with Synapse and assist in efficient querying.

An interactive dashboard is then created using power BI. It offers such insights as the sales performance and customer demographics. The data are easily explored using filters.

2.4 Automation and Monitoring

Azure Data Factory triggers enable the pipeline to be automated, which enables it to operate on a daily basis without human intervention. ADF and Synapse monitor tools are employed to make sure that the pipeline is running smoothly and efficiently.

2.5 Security

Azure Key Vault and Azure Entra ID (RBAC) are used to implement security. The sensitive information is stored in Azure Key Vault and user access to the system is regulated by RBAC.

3.0 Technology Used

Technology	Pipeline Stage	Function	How it Supports the Pipeline
SQL Server (On-Prem)	Data Source	Stores raw customer and sales data	Serves as the point of entry of the pipeline and provides data to be ingested into the Azure environment
Azure Data Factory (ADF)	Data Ingestion & Orchestration	Extracts data from SQL Server and loads it into Azure Data Lake. Also schedules and manages workflow	Facilitates automatic data flow and guiding the pipeline from source to the storage without any difficulties
Azure Data Lake Storage Gen2 (ADLS Gen2)	Data Storage	Stores data in Bronze (raw), Silver (cleaned), and Gold (processed) layers	Can be used as a centralized storage system, which facilitates various steps of data processing in the pipeline
Azure Databricks	Data Transformation	Cleans, processes, and transforms data using Apache Spark	Transforms raw data into high quality data that can be analyzed
Azure Synapse Analytics	Data Warehousing	Stores processed data and provides structured views for querying	Facilitates effective querying and sets data ready to report and analyze
Microsoft Power BI	Data Visualization	Creates dashboards and visual reports based on processed data	Converts data into useful information for decision-making
Azure Key Vault	Security	Stores sensitive credentials securely	Guarantees the security of pipeline data and connections
Azure Entra ID (RBAC)	Access Control	Manages user roles and permissions	Provides the safe control of access throughout the pipeline

4.0 Data Pipeline Architecture

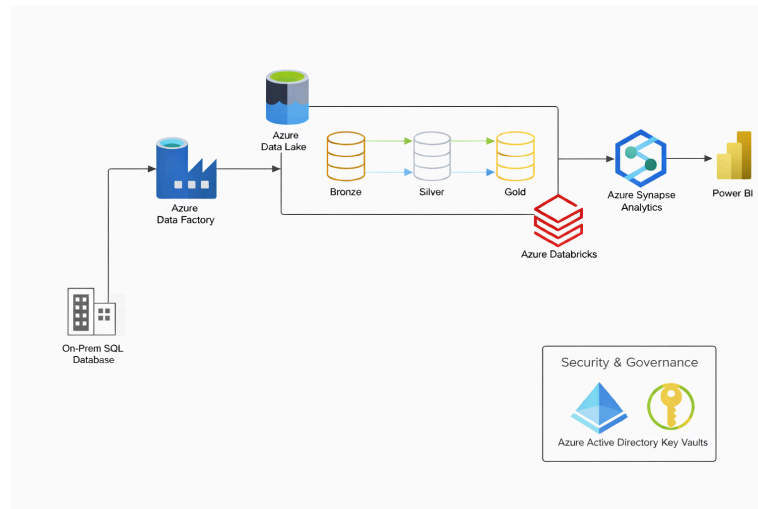


Figure 4.1: End-to end Data Pipeline Architecture

The data pipeline architecture of this project shows the end-to-end data flow from the source system to the ultimate visualization layer. The pipeline starts with a local SQL server database that will act as the main source of data as the starting point of the pipeline with raw customer and sales information.

Azure Data factory (ADF) is used to extract the data, carrying out the process of data ingestion and coordinating the workflow. ADF moves the data to Azure Data Lake Storage Gen2 (ADLS Gen2) and the data is stored in a structured layered format consisting of Bronze, Silver, and Gold layers.

Data transformation is done in the data lake of Azure databricks. Raw data is stored in their original form in the Bronze layer. The Silver layer holds purged and processed data and the Gold layer holds refined, business-ready data that can be used in analytics.

Once transformed, the processed data is then loaded into Azure Synapse Analytics where structured views are then created to facilitate querying and analysis. Lastly, Microsoft Power BI links with processed data to create interactive visualizations and dashboards to gain business insights.

Furthermore, access control is implemented with the help of the Azure Entra ID (RBAC) and secure management of sensitive credentials with the help of Azure Key Vault. This will guarantee safe and efficient operation of the whole pipeline.

5.0 MS Power BI dashboard

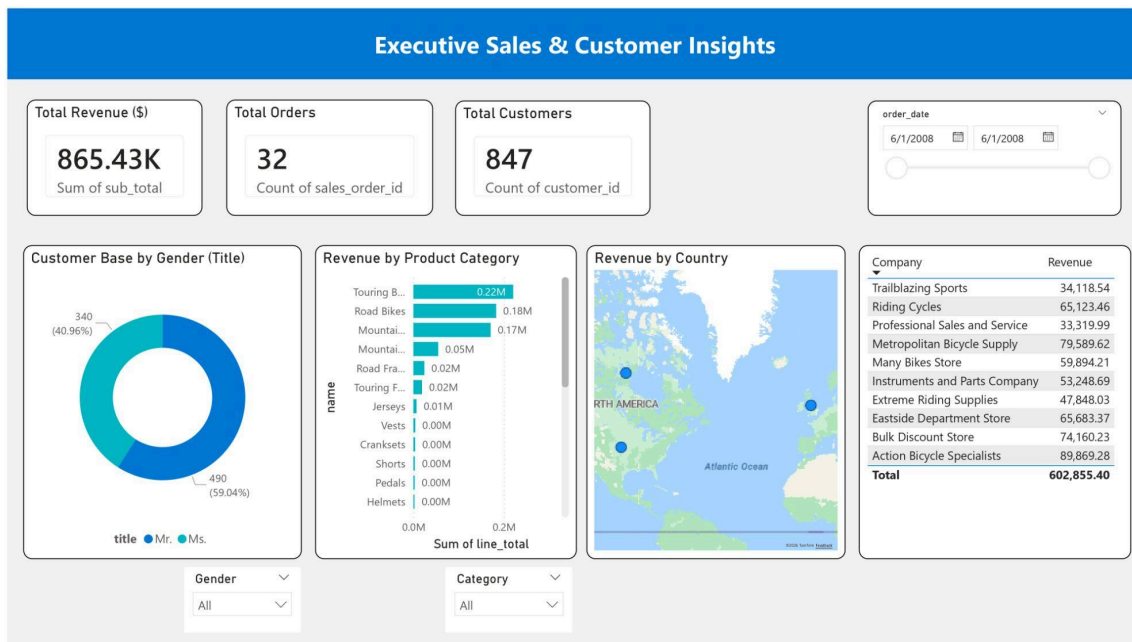


Figure 5.1: Power BI Dashboard of Executive Sales & Customer Insights

This Power BI dashboard connects to Azure Databricks, which was used to clean and transform the raw SQL data into a "Gold" layer for analysis. This automated integration ensures that the Power BI visuals are powered by accurate, business-ready data delivered through a secure pipeline.

This Executive Sales & Customer Insights dashboard shows multiple visuals, each visual was chosen for a specific business use. KPI cards provide an instant summary of revenue and order totals, while the Donut Chart tracks customer gender to help with targeted marketing. The Bar Chart identifies which product categories drive the most profit, the Map shows where sales are geographically concentrated, and the Table lists the top 10 customers to help manage important business relationships.

6.0 Reflection

LEE YIN SHEN	From this tutorial, I gained hands-on experience to build a complete Azure pipeline, moving data from a local SQL database into Power BI. I used Azure Data Factory for ingestion, Databricks for cleaning data, and Synapse to serve the final views. I also learned how to use Power BI to present complex data through professional and clear visualizations. Even though I followed the tutorial, I still faced several challenges like authorization, server connections, and tool integration. I had to troubleshoot these errors myself to solve them. This project also taught me to manage credentials securely using Azure Key Vault and to automate the entire data lifecycle.
JOANNE CHING YIN XUAN	Through this project, I learned how to use Azure tools to build a data pipeline and connect different services like Data Factory, Databricks, Synapse, and Power BI. During the process, I faced many issues that required me to recheck every step, especially with errors like permission errors, configuration problems, schema inference issues and wrong file paths. Besides, the YouTube tutorial was an older version of Azure, many steps were different from the current interface. I could not fully rely on the video and had to search for solutions, and test different approaches. Overall, this experience helped me improve my troubleshooting skills and be more patient and independent when solving technical problems.
EVELYN GOH YUAN QI	This project provided me with a great practical experience in creating an end-to-end data engineering pipeline with Microsoft Azure. I also

	got to understand the interaction of various tools including Azure Data Factory, Data Lake, Databricks, Synapse, and Power BI in processing raw data to valuable insights. This made me have a better idea of how a real-world data pipeline works. Some of the challenges encountered during the implementation process were with the configuration of services and troubleshooting problems associated with permissions and connections. To address such problems, I looked for other solutions, experimented with various settings, and used online materials, which enhanced my troubleshooting skills. Working in a group helped me to understand the value of communication and teamwork because we supported one another to solve problems and accomplish tasks effectively. Overall, this project improved my technical skills and my confidence working with cloud-based data engineering solutions.
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